Jaren Lim

+65 9826 8686 | jarenlim100@gmail.com | https://github.com/jarenlimhaan

Experience

Backend Engineer (SRE / Plaform Engineering) Jul 2026 – Nov 2026
NTT DATA *Singapore*

- Operated Kubernetes-based applications across development and staging environments, using kubectl, Kubernetes events, logs, and metrics to troubleshoot pod failures, failed rollouts, resource constraints, and service connectivity issues.
- Developed and optimized GitLab CI/CD pipelines for Kubernetes deployments, reducing deployment time by over 70% through automated build and deployment workflows.
- Monitored application and infrastructure health using Prometheus, Grafana, and Datadog, and analysed centralized ELK logs to investigate service and deployment issues.
- Developed Express.js backend services and REST APIs for task management, workflow automation, and project tracking.

GenAI Developer May 2026 – Jul 2026
KPMG *Singapore*

- Built AI-powered document processing workflows using Microsoft Copilot Studio and Azure AI Foundry to automate invoice extraction, review, and validation.
- Developed multi-agent workflows for document parsing, validation, reasoning, and structured output generation, reducing invoice processing time from 4–5 hours to approximately 10 minutes.
- Developed Python and LangChain components to implement business-specific validation logic and integrate it into automated document-processing workflows.

Software Engineer May 2023 – Aug 2025
RSAF Agile Innovation Digital (RAID) *Singapore*

- Developed NestJS backend services and REST APIs for a constraint-based duty scheduling platform, implementing automated roster generation and schedule management workflows.
- Implemented scheduling logic for shift coverage, leave conflicts, personnel qualifications, and fairness constraints to automatically generate feasible duty allocations.
- Built Datadog dashboards and monitors using application and infrastructure metrics to detect service degradation and support production incident investigation.
- Developed Next.js frontend workflows for creating, reviewing, and modifying generated schedules, integrating them with NestJS backend APIs.

Machine Learning Engineer Sep 2021 – Feb 2022
SAP *Singapore*

- Improved recommendation performance by approximately 5% through feature engineering, embedding optimization, and iterative model tuning.
- Developed and evaluated a hybrid recommendation system combining GPT-2 with neural collaborative filtering models in Python to improve recommendation ranking quality.

Projects

ESMOS Healthcare Portal (Dell award Finalist) | *AWS, Terraform, EKS, ArgoCD, Helm* Apr 2026

- Provisioned AWS infrastructure using Terraform across EKS, RDS, EFS, VPC networking, WAF, CloudTrail, and automated backups.
- Implemented GitOps-based Kubernetes deployments using ArgoCD and Helm, with GitHub Actions integrating Snyk, Checkov, kube-linter, and kubeconform for security and configuration validation.

Lumina (Open Government Project - The Good Hack) | *React, FastAPI, Kafka, Twilio, SEA-LION* May 2026

- Built an AI-powered coordination platform for Medical Escort and Transport (MET) bookings, reducing manual administrative workflows by over 40%.
- Architected an event-driven microservices backend using Python, FastAPI, and Kafka to automate subsidy eligibility checks, provider dispatch, and workflow orchestration.
- Developed a multilingual AI voice assistant using Twilio and SEA-LION to automate user support across 7 Southeast Asian languages; selected for incubation support by Open Government Products.

Technical Skills

Languages & Frameworks: Python, Go, Java, JavaScript/TypeScript, React, Next.js, NestJS, Express.js, FastAPI
Cloud & Infrastructure: AWS, Kubernetes, Docker, Terraform, Helm, ArgoCD, Linux
DevOps & Observability: GitLab CI/CD, GitHub Actions, GitOps, Prometheus, Grafana, Datadog, ELK, Loki, Tempo
Databases: PostgreSQL, MongoDB, SQLite
Certification: AWS Certified Solutions Architect – Associate

Education

Singapore Management University Aug 2024 – Sep 2027
Bachelor of Computing in Information Systems (Magna Cum Laude)